



SMART LIGHT

bricks 4 kidz

QUESTION

What do we usually need to do when it's **dark**?



QUESTION



Turn on light

QUESTION



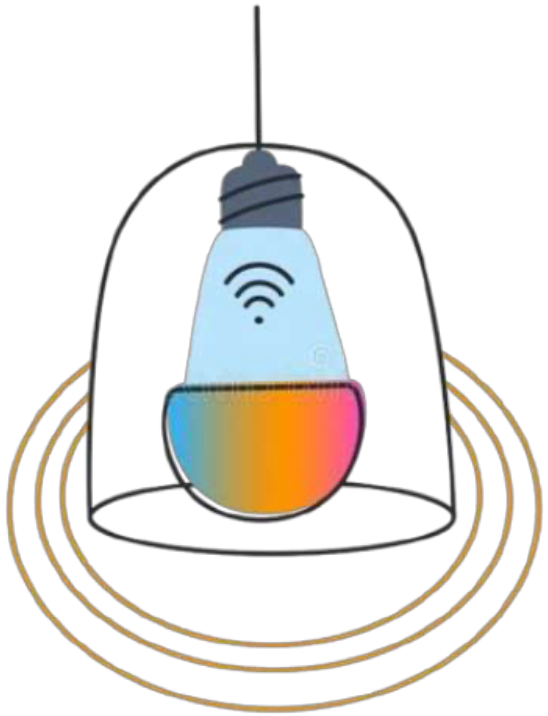
We use a switch to turn the light on and off.
But what if we can't find it?

QUESTION



We need an automatic light

QUESTION



This light will turn on when it’s dark
Turn off when it’s bright

LESSON FLOW

SMART LIGHT



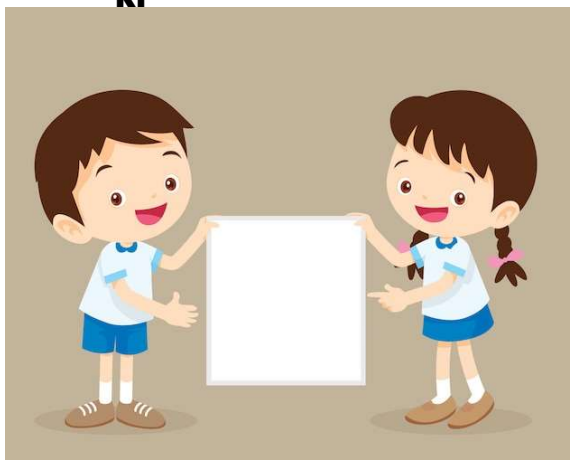
TEMPERATURE SENSOR



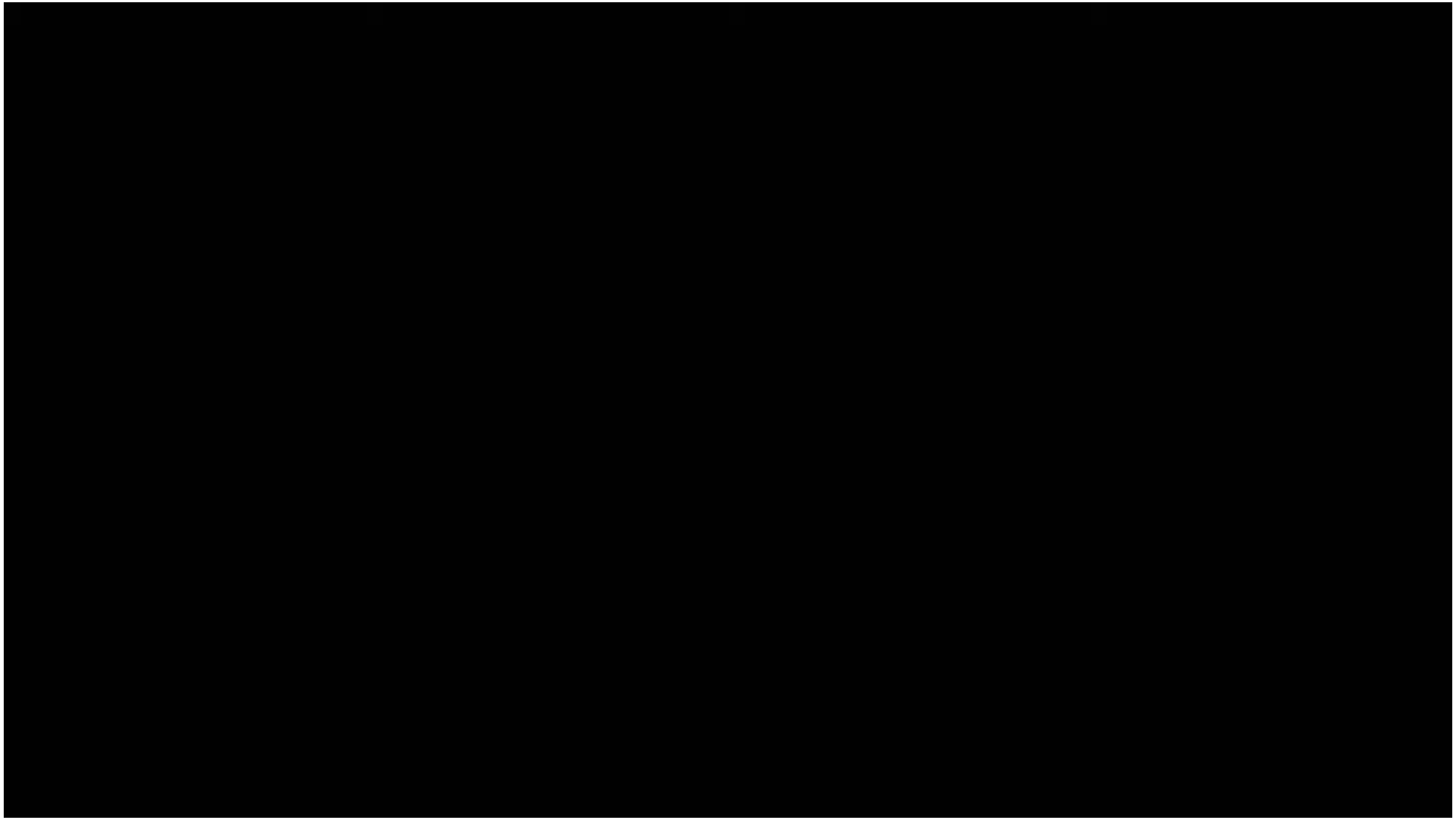
MODEL



PRESENTATIO N

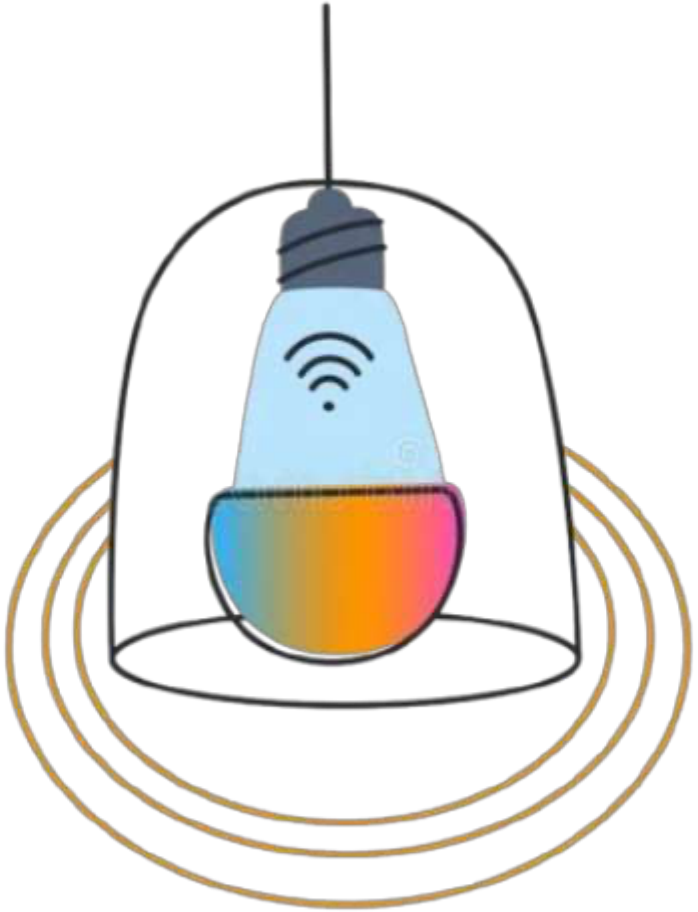


SMART LIGHT



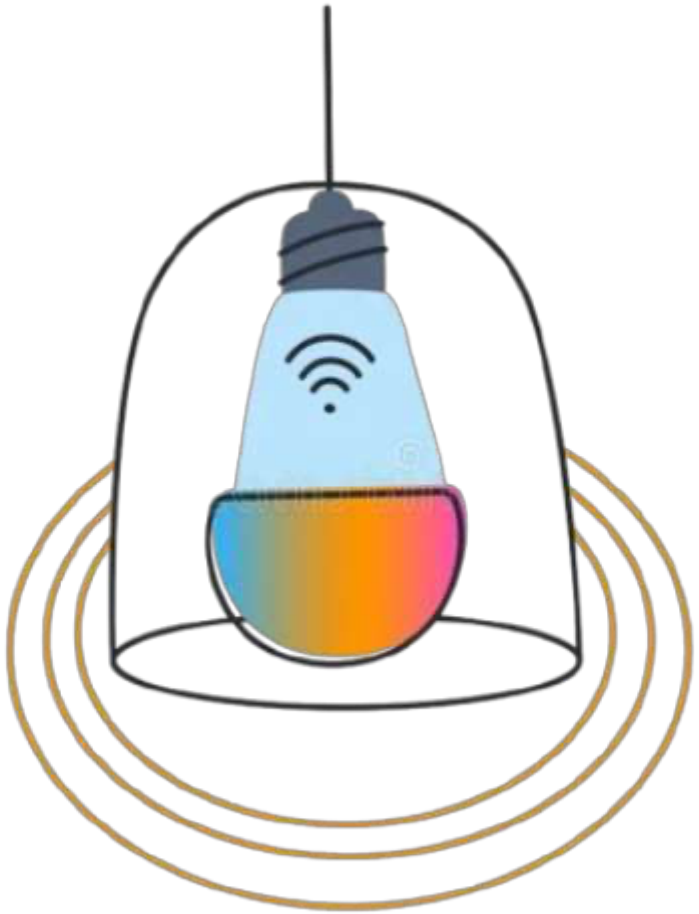
SMART LIGHT

What is a automatic/smart light?



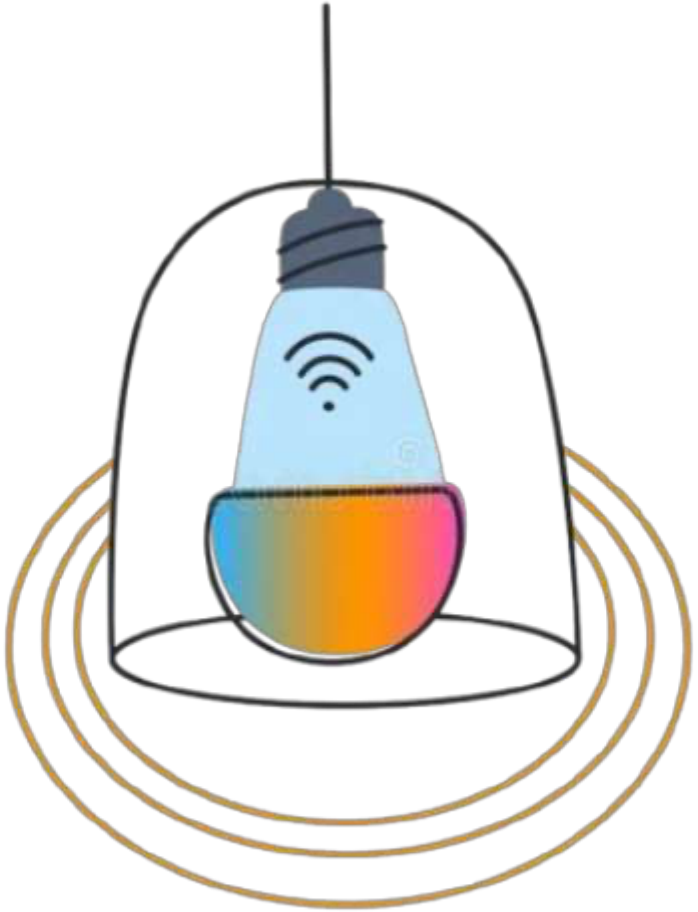
SMART LIGHT

An **automatic light** is a type of light that can turn **on** or **off** without direct human control, thanks to **sensors** or smart **controllers**



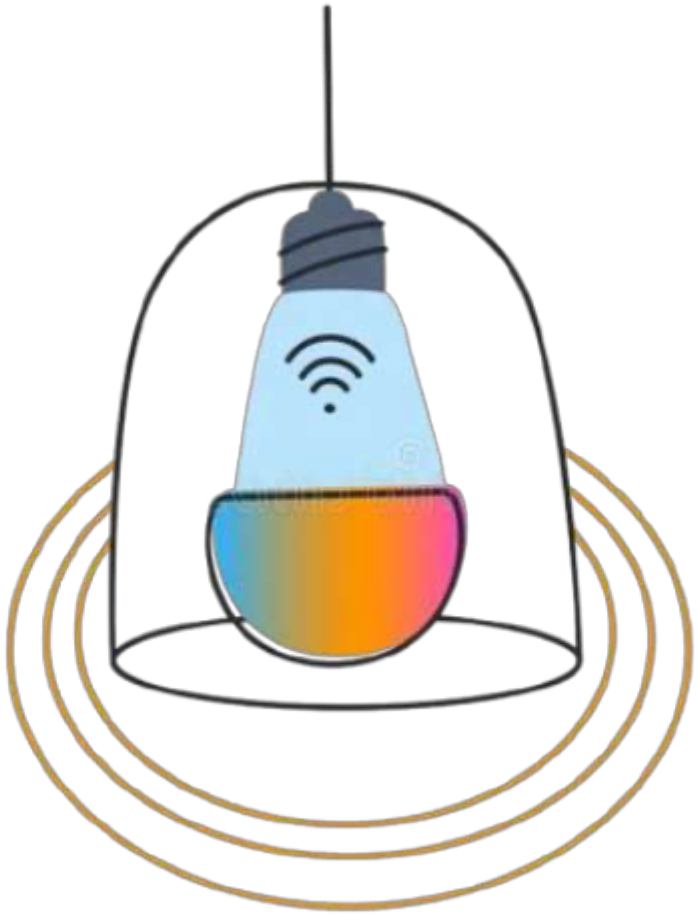
SMART LIGHT

How are smart lights useful?

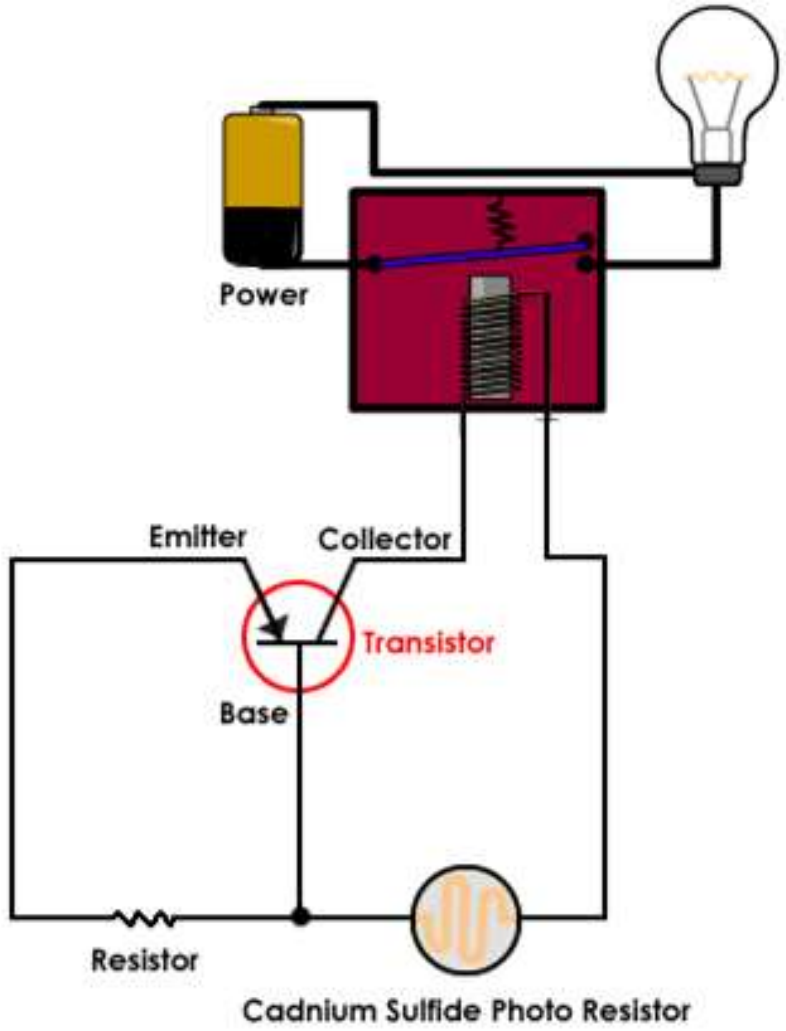


SMART LIGHT

- Save electricity
- Convenient, no need for manual switching
- Increase safety (turn on the light when someone enters).



SMART LIGHT

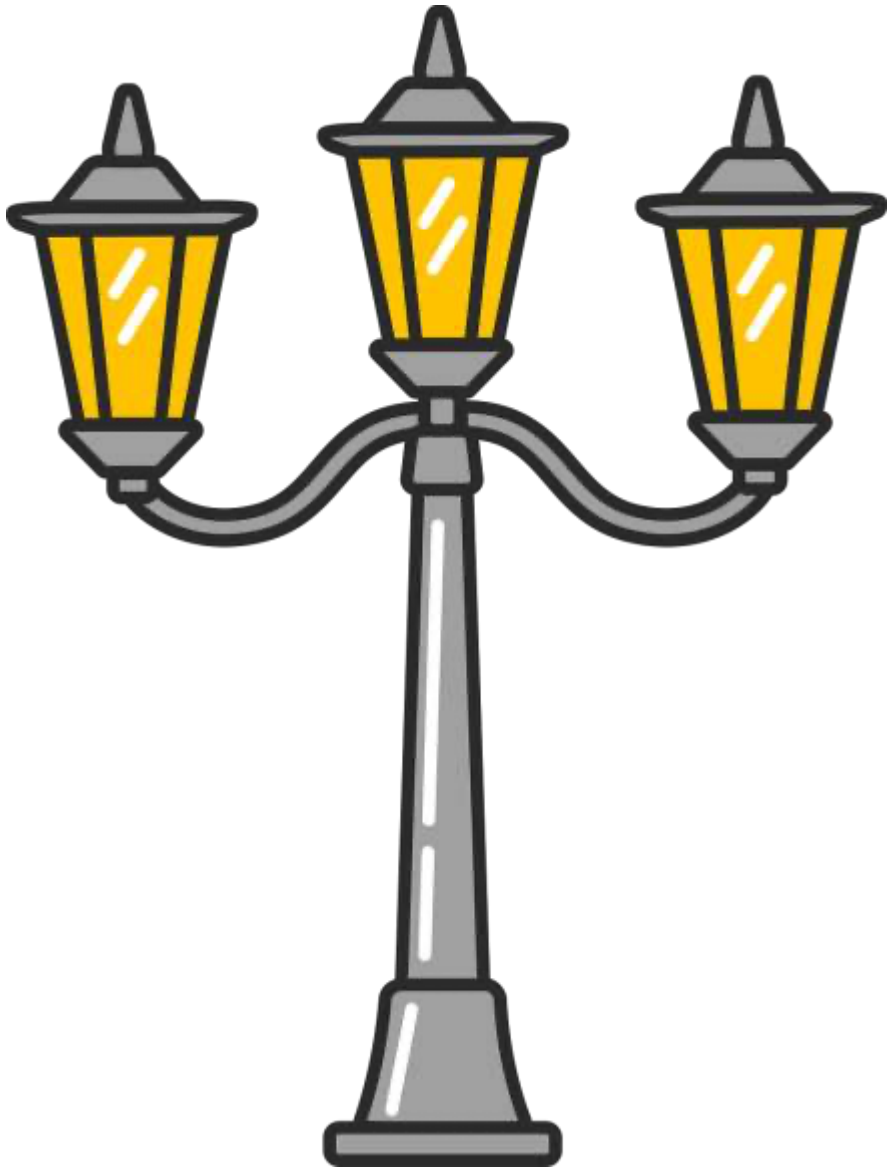

Power
Emitter
Collector
Transistor
Base
Resistor
Cadnium Sulfide Photo Resistor

SMART LIGHT

Where can you see the smart light?



SMART LIGHT



Street Light

SMART LIGHT

Smart house



LIGHT SENSOR

Why do smart light know when it’s dark/bright?

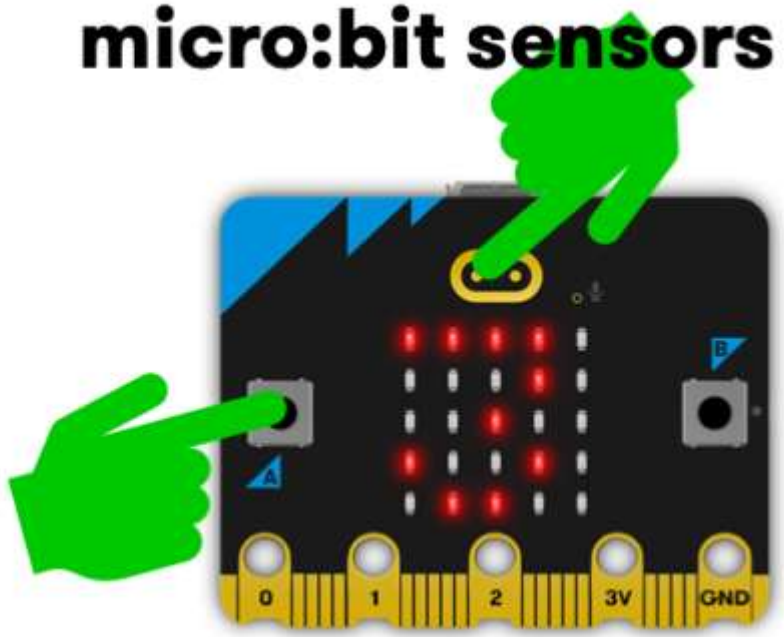


LIGHT SENSOR

A light sensor is a device that can **detect** and **measure** the intensity of **light** in the surrounding environment.



LIGHT SENSOR



Light sensor in micro:bit

LIGHT SENSOR



Guess how a light sensor works.

LIGHT SENSOR

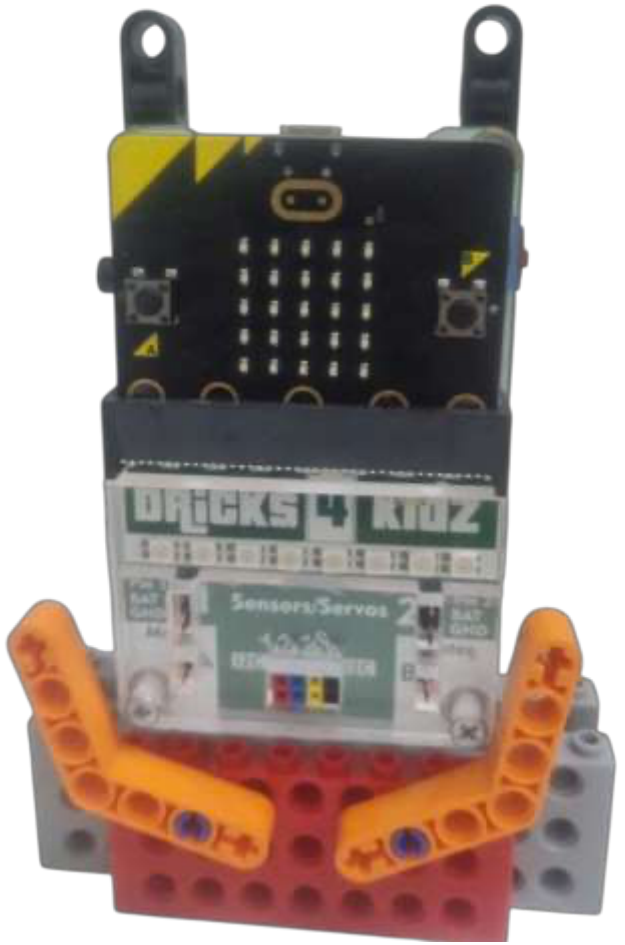


- When the light is **strong** → the sensor receives a **high** value.
- When the light is **weak** or **dark** → the sensor receives a **low** value.

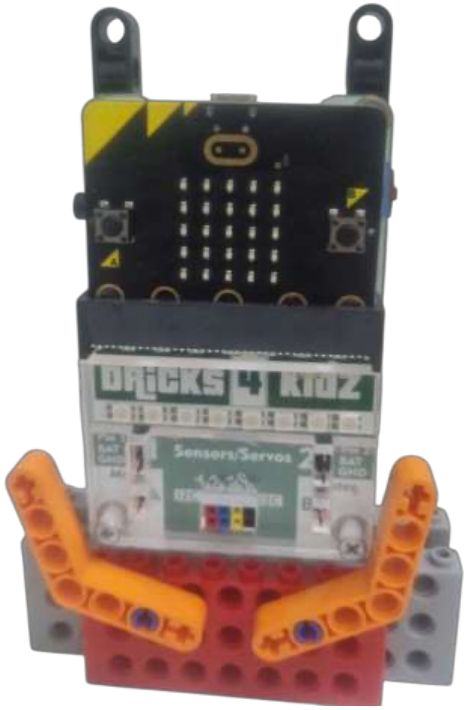

LIGHT
SENSING

SMART LIGHT

LET’S BUILD A SMART LIGHT

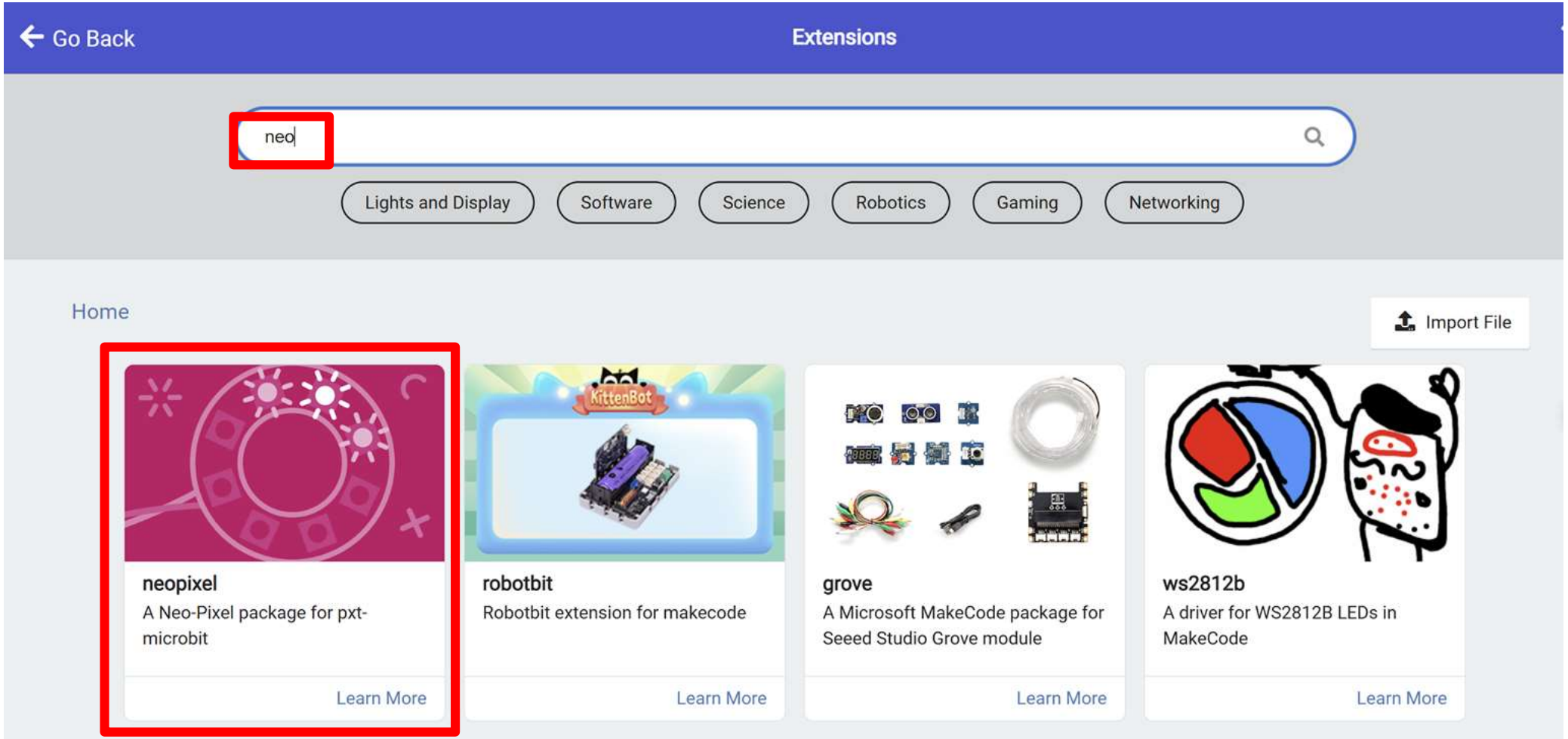


SMART LIGHT



Turn on the Neo Pixel Light

SMART LIGHT


Go Back
Extensions
neo
Lights and Display
Software
Science
Robotics
Gaming
Networking
Home
Import File
neopixel
A Neo-Pixel package for pxt-microbit
Learn More
KittenBot
robotbit
Robotbit extension for makecode
Learn More
grove
A Microsoft MakeCode package for Seeed Studio Grove module
Learn More
ws2812b
A driver for WS2812B LEDs in MakeCode
Learn More

SMART LIGHT



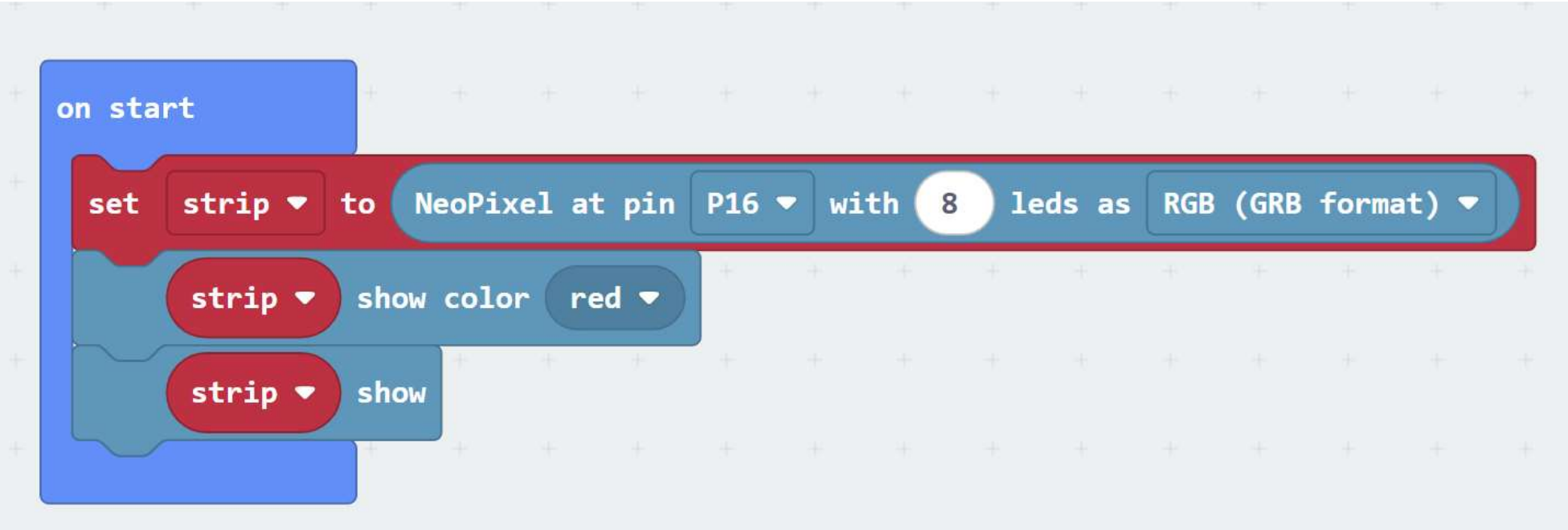
Set up the Neo Pixel Light in board

SMART LIGHT



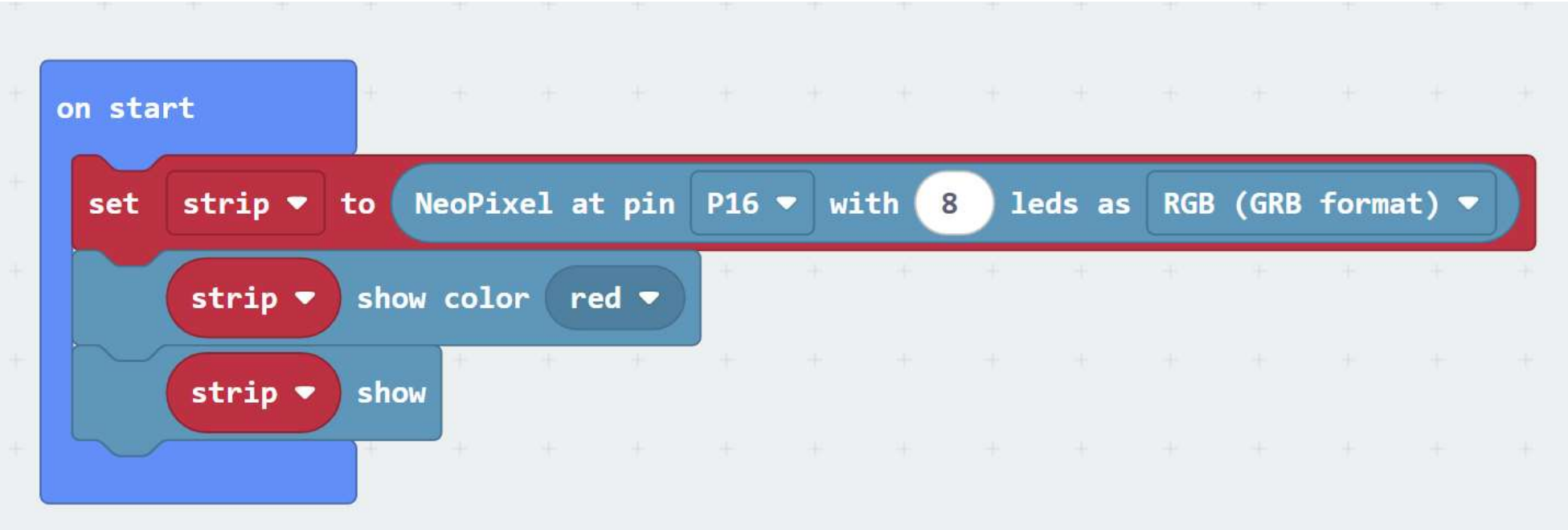
Set up the Neo Pixel Light in board

SMART LIGHT



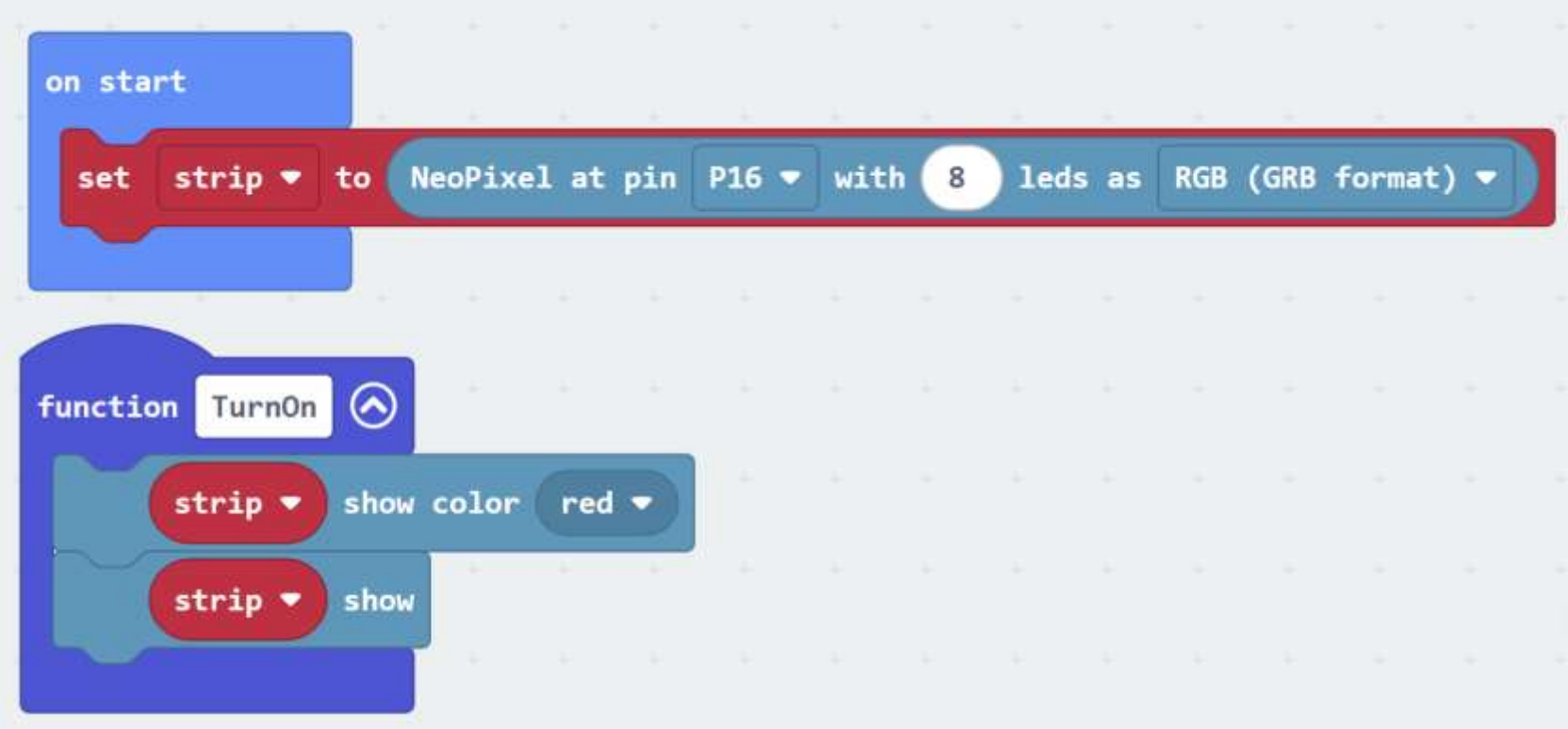
Turn on the neo pixel light

SMART LIGHT



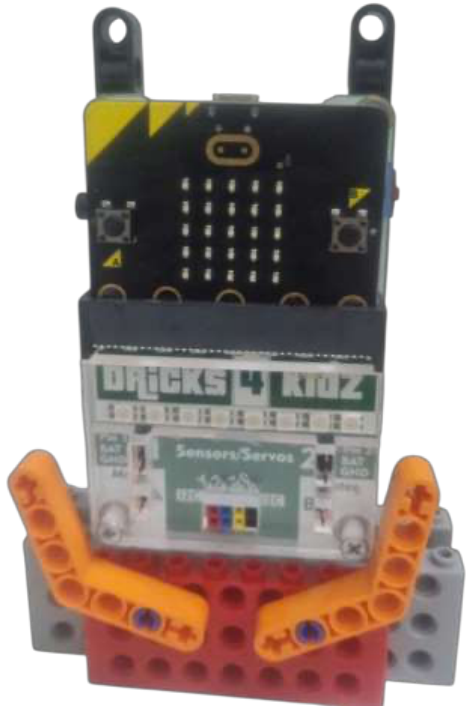
Make a function for turn on light

SMART LIGHT



Make a function for turn **on** light

SMART LIGHT



Read the value of light sensor

SMART LIGHT



Read the value of light sensor

SMART LIGHT



If it's **light**, turn **on** the light
If it's **dark**, turn **off** the light

SMART LIGHT

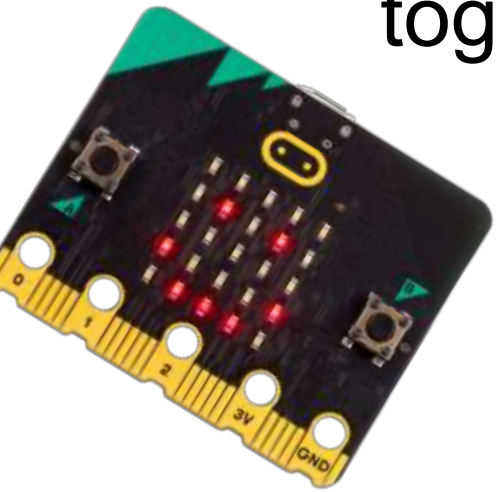


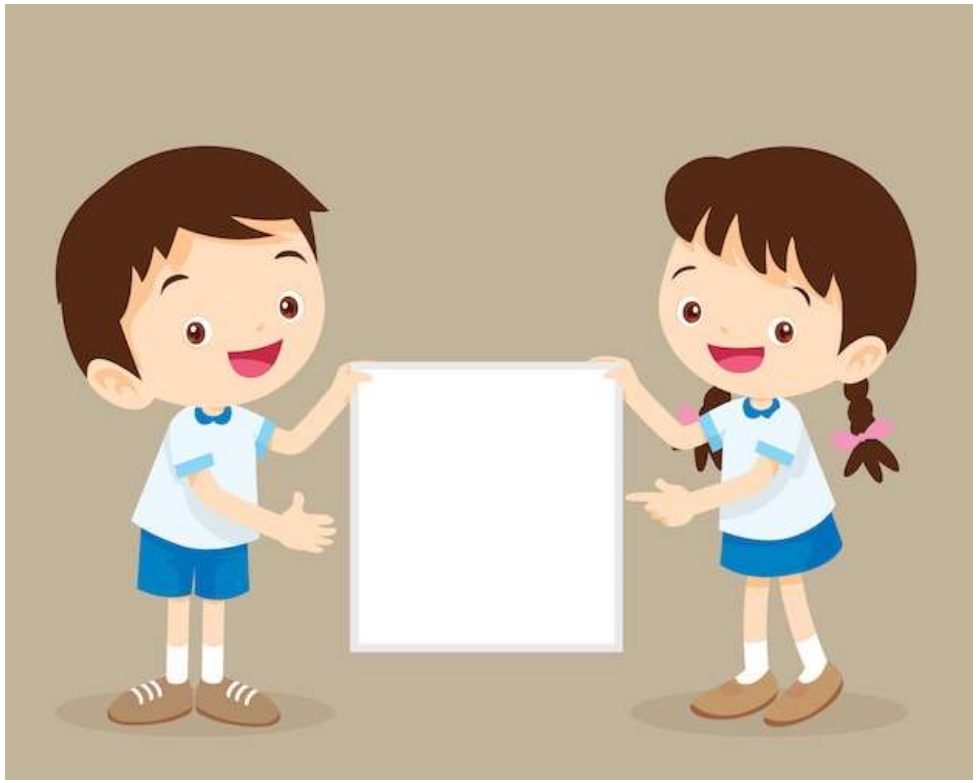
If it's **light**, turn **on** the light
If it's **dark**, turn **off** the light



PRESENTATION

Let's introduce our product to everyone together!



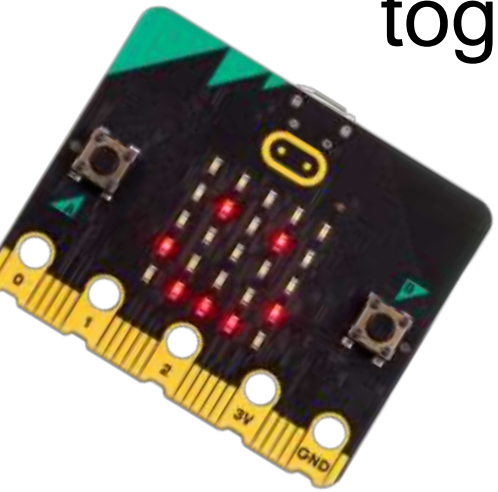






PRESENTATION

Let's introduce our product to everyone together!



What is your model's name?

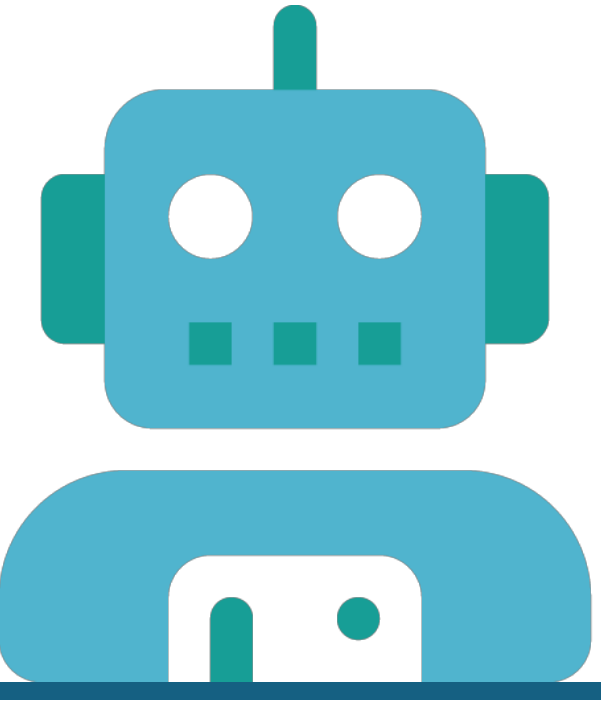
" – My system is smart light…"



How do it work?

"It will turn on when…"

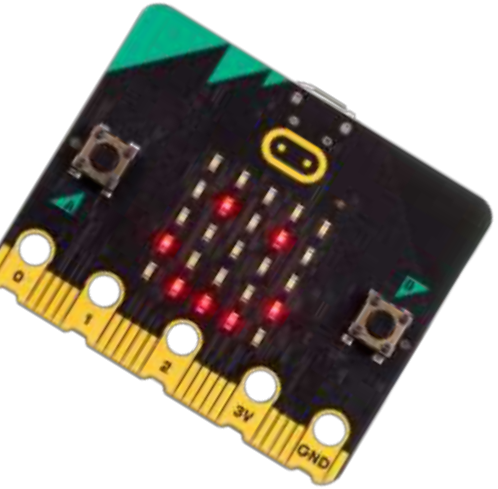
TIME TO CLEAN UP!!!







SUMMARY



SMART LIGHT

LIGHT SENSOR



NEOPIXEL LIGHT


THANK
YOU!

